

Caroline Schwengler

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EDUCATION

Master of Science, *University of Arizona*

Expected May 2027

Major: Statistics and Data Science

Bachelor of Arts, *University of Arizona*

2026

Majors: Statistics and Data Science, Mathematics, French Studies

GPA: 3.86

Minor: Computer Science

SKILLS

Languages

Python, SQL, R, Java, MATLAB.

ML

PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib, Streamlit.

Cloud/Data

AWS, Snowflake, Git.

Coursework

Machine learning, mathematics of AI, experimental design, statistical consulting.

WORK EXPERIENCE

Data Science Intern

May 2025 — Aug 2026

World Wide Technology

Denver, Colorado

- Evaluating over 10,000 historical AI chatbot queries to quantify trends in accuracy, latency, user churn and retention and created visualizations for a technical and non-technical audience.
- Applied clustering to identify user cohorts and characterize behaviors driving high and long-term usage.
- Built Snowflake stored procedures to classify query topics and translate Spanish queries to English, improving data interpretability and processing.
- Developed 10 prioritized recommendations to improve user experience, model accuracy, and latency.

Course Coordinator - Undergraduate Teaching Assistant

Aug 2023 — May 2026

University of Arizona

Tucson, Arizona

- Led weekly meetings of approximately 10 teaching assistants and supported course instruction and administration.
 - Hosted office hours and supported instruction for undergraduate mathematics and computer science courses, assisting students with programming and quantitative problem solving.
 - Graded assignments, provided individualized feedback, and delivered review lectures to classes of approximately 20 students twice weekly.
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PROJECTS

Comparative Analysis of Protein Prediction Models

- Investigated protein-structure prediction models including AlphaFold2, Boltz, and Chai using Python, evaluating their capabilities and applicability to research workflows.
- Developed concurrent model-execution framework in Python and created a user interface using Streamlit for protein-structure predictions.
- Benchmarked models across GPU requirements, cost, runtime, and performance.

Prediction of Failure after Kidney Transplantation Surgery

- Developed predictive models for post-transplant failure using clinical data, comparing logistic regression, segmented regression, stepwise regression, and K-nearest neighbors by AUC.
- Modeled minute-by-minute ventilation data using segmented regressions and spline models to characterize respiratory trends across patients.
- Presented research and animated visualizations at two biostatistics and mathematics research conferences.